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Electric load forecasting with recency effect: A big data approach

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ABSTRACT

Temperature plays a key role in driving the electricity demand. We adopt the "recency effect", a term drawn from psychology, to represent the fact that the electricity demand is affected by the temperatures of the preceding hours. In the load forecasting literature, the temperature variables are often constructed in the form of lagged hourly temperatures and moving average temperatures. In the past, computing power has limited the amount of temperature variables that can be used in a load forecasting model. In this paper, we present a comprehensive study to model the recency effect using a big data approach. We take advantage of modern computing power to answer a fundamental question: *how many lagged hourly temperatures and/or moving average temperatures are needed in a regression model in order to capture the recency effect fully without compromising the forecasting accuracy?* Using a case study based on data from the load forecasting track of the Global Energy Forecasting Competition 2012, we first demonstrate that a model with the recency effect outperforms its counterpart (a.k.a. Tao's Vanilla Benchmark Model) by 18% to 21% for forecasting the load series at the top (aggregated) level. We then model the recency effect in order to customize load forecasting models at the bottom level of a geographic hierarchy, again showing a superiority over the benchmark model of 12% to 15% on average. Finally, we discuss four different implementations of the recency effect modeling by hour of a day. In addition, this paper also presents two interesting findings: 1) the naive models are not useful for benchmark purposes in load forecasting at aggregated level due to their lack of accuracy; and 2) slicing the data into 24 pieces to develop one model for each hour is not necessarily better than building one interaction regression model using all 24 hours together.
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1. Introduction

At the inception of electric power systems, lighting was the only end use of electricity, meaning that the electricity demand was driven primarily by calendar variables. As more and more electricity-powered appliances were invented, the end use became diversified. The increasing penetration of electrical air conditioning systems made the role of weather more and more important in driving the electricity demand. Ever since the 1940s, people have realized that the electric load depends strongly on the weather (Dryar, 1944).

In the pre-PC (personal computer) era, utility planners and operators created lookup tables and charts, based on historical data and past experience, for capturing the relationship between the load and weather variables such as temperature and humidity. They then forecasted the load using these charts and tables, together with rulers and intuitions (Hong, 2014).

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When people started using computers for load forecasting in 1980s, the computing power was very limited. Quite often, the model building procedures for selecting variables and estimating parameters had to be conducted offline, meaning that the computer was performing online calculations of the load forecast based on the new data and previously calculated variables and parameters (Gross & Galiana, 1987). The offline model building scheme meant that the model(s) could not be updated in real time to reflect the most recent status of the power system. As a consequence, the forecasting accuracy was more or less compromised.

The technological advancement through the late 1990s quickly eliminated the need for offline computation for many load forecasting techniques. The benefits of the increase in computing capability also meant that people started to apply some more computationally intensive techniques such as artificial neural networks (Hippert, Pedreira, & Souza, 2001) and autoregressive integrated moving average models (Weron, 2006) to load forecasting. At the same time, people also started using large numbers of variables in load forecasting models (Hippert et al., 2001).

In psychology, the recency effect refers to the fact that human beings tend to remember the most recent items. The power grid is similar, in that its demand tends to be affected significantly by the recent temperatures. Hong (2010) was the first to adopt this term for illustrating part of a systematic load forecasting methodology, which used lagged temperatures to enhance the load forecasting accuracy of a benchmark model. Since then, this term "recency effect" has been accepted widely in the US utility industry, and has become part of a commercial software package (SAS® Energy Forecasting) that is currently being used by many power companies worldwide. Note that many papers in the load forecasting literature have reported the use of lagged temperatures. Papalexopoulos and Hesterberg (1990) used lagged temperatures to calculate lagged heating and cooling degree days for regression models. One winning team (Ben Taieb & Hyndman, 2014) of the Global Energy Forecasting Competition 2012 (GEFCom2012) used lagged hourly temperature and average daily temperature variables in the competition. Another winning team of GEFCom2012 (Nedellec, Cugliari, & Goude, 2014) used exponentially smoothed temperature variables. Nevertheless, there has never been an in-depth study of different model sizes, to investigate whether large numbers of lagged and average temperature variables can help to improve the forecast accuracy.

Despite the big improvement on the computation side, model building can still take significant amount of time if one wants to test many variables. Indeed, at times people still have to juggle the tradeoff between the frequency of model updates and the sufficiency of the variables. As was discussed by Hong (2010), for example, lagged temperature variables were limited to the past three hours of temperatures due to computational constraints. The limitation of computing power constraints was also an issue in earlier decades, as was discussed by Gross and Galiana (1987).

The methodology presented in this paper is a continuation and extension of the work of Hong (2010). We attempt to take advantage of modern computing power in order to answer a fundamental question:

How many lagged hourly temperatures and/or daily moving average temperatures are needed in a regression model so as to capture the recency effect fully without compromising the forecasting accuracy?

Note that this paper does not explore exponentially weighted temperature variables, primarily in order to avoid discussing algorithms for fine-tuning the exponential weights. For instance, one heuristic method for selecting the base for the exponential weights was discussed by Hong (2010). Other notable discussions of exponential smoothing for electric load forecasting are provided by Taylor and McSharry (2007) and Weron (2006). The lagged temperatures and moving average temperatures covered in this paper can be regarded as a typical representation of "recency", in the sense that we assign a weight of one to each observation in the moving window and a weight of zero to observations outside the moving window. Nevertheless, the proposed framework does not exclude the use of exponentially weighted temperatures.

Following Hong, Wang, and White (2015), we develop a case study based on the load forecasting data from GEFCom2012 published by Hong, Pinson, and Fan (2014a). There are two big-data aspects of this paper: (1) we customize the model for each zone of a geographic hierarchy and each hour of the day; and (2) we take advantage of modern computing power to develop large load forecasting models with thousands of variables. Section 4.2 provides further discussion about big data in load forecasting.

This paper makes the following significant contributions to the load forecasting literature: (1) this is the first comprehensive study on the modeling of the recency effect without computational constraints; (2) this is the first time that the recency effect has been applied to hierarchical load forecasting, in the context of both geographical and temporal hierarchies, where the recency effect is being modeled in a customized fashion for each zone and each hour of a day; and (3) publicly available data are used to conduct the case study, so that future researchers can reproduce our results.

2. Background

2.1. Data description

One of the objectives of GEFCom2012 was to establish a benchmarking data pool to make it easy for researchers in the energy forecasting community to compare models. In this paper, we use the data from the hierarchical load forecasting track of GEFCom2012, which includes 4.5 years of hourly load and temperature across 21 zones (Z_i, $i = 1, 2, \ldots$) of a US utility, of which Z_{21} was the sum of the first 20 zones (Hong, Pinson et al., 2014a). We use the first 4 years of load and temperature data in this paper.

As per the case study by Hong et al. (2015), we are conducting out-of-sample tests instead of selecting models based on in-sample fits. Here, we slice the data to three pieces, the first two years (2004–2005) for training (or in-sample fit, for parameter estimation), the next year (2006) for validation (or post-sample fit, for model selection), and the last full calendar year (2007) for testing (or out-of-sample testing, for a summary of error statistics). The four years of Z_{21} load data are shown in Fig. 1.

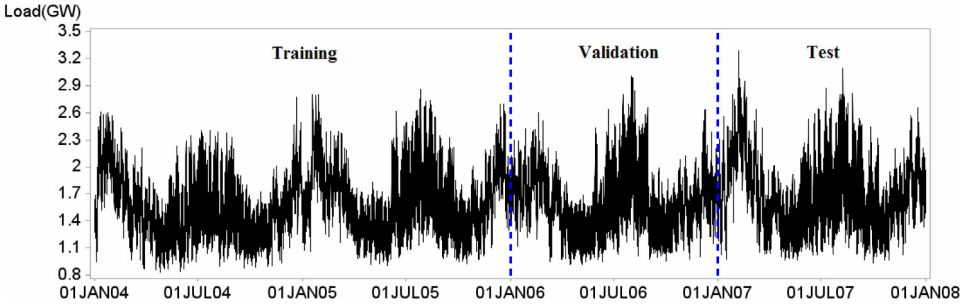

Fig. 1. Data from the GEFCom2012 load forecasting track.

We also use the weather station selection and load forecasting results of Hong et al. (2015) as a benchmark for the methodology proposed in this paper. In other words, actual temperature observations are used across the forecast horizons, which is widely done in the industry when developing load forecasting models, as was discussed by Nedellec et al. (2014). A more in-depth discussion on the use of weather variables in ex ante vs. ex post load forecasting was presented by Hong et al. (2015).

2.2. Tao's vanilla benchmark model

Tao's vanilla benchmark model was first proposed by Hong (2010), then used to produce benchmark scores for GEFCom2012 (Hong, Pinson et al., 2014a). In GEFCom2012, this model was ranked in the top 25% among over 100 teams. It has also been implemented as a base model in the commercial software package SAS Energy Forecasting. The same model was used in the weather station selection framework proposed by Hong et al. (2015). The model is a multiple linear regression model:

$$y_t = \beta_0 + \beta_1 \, Trend_t + \beta_2 M_t + \beta_3 W_t + \beta_4 H_t + \beta_5 W_t H_t + f(T_t), \quad (1)$$

where y_t is the load forecast for time (hour) t; β_i are the coefficients estimated using the ordinary least squares method; M_t, W_t and H_t are the month-of-the-year, day-of-the-week, and hour-of-the-day classification variables corresponding to time t, respectively; T_t is the temperature at time t; and

$$f(T_t) = \beta_6 T_t + \beta_7 T_t^2 + \beta_8 T_t^3 + \beta_9 T_t M_t + \beta_{10} T_t^2 M_t + \beta_{11} T_t^3 M_t + \beta_{12} T_t H_t + \beta_{13} T_t^2 H_t + \beta_{14} T_t^3 H_t. \quad (2)$$

This model, and those to be discussed in later sections, use temperature and calendar variables only, not past loads, mainly because many jurisdictions do not allow past loads to be used for long term load forecasting, in order to avoid reducing the interpretability of the model. Due to the simplicity of regression analysis and its wide adoption for load forecasting, we used multiple linear regression models to convey the ideas of recency effect modeling. Nevertheless, the recency effect can also be modeled using other techniques, such as neural networks and support vector regression models.

2.3. Missing salient features

Using two years of data (2005–2006) to estimate the parameters of the benchmark model, we can perform one-year-ahead ex post forecasting for the 8760 hours in the year 2007. Fig. 2 shows the actual and forecasted loads from Tao's vanilla benchmark model, overlaid with temperature, for a summer week and a winter week. We observe the following discrepancies between the forecast and actual loads:

(1) The benchmark model over-forecasts the summer peaks for consecutive days (6/22–24) and under-forecasts the winter peaks for consecutive days (11/15–20);
(2) The benchmark model under-forecasts the troughs for consecutive days (11/16–20);
(3) The forecast from the benchmark model leads or lags the actual load for consecutive hours on several days (6/20, 6/23, 6/24, 11/16–19).

In addition, we also observe a lag between the temperature and load profiles in the summer, which indicates that the absence of temperatures from preceding hours in the load forecasting model may be the reason for some of the discrepancies mentioned above. This further motivates our study of the recency effect. The same two weeks will be plotted later in Fig. 4 to show the improvement achieved by modeling the recency effect.

3. Recency effect

3.1. At an aggregated level

We capture the missing salient features discussed above by including the temperatures of preceding hours in the model. In this paper, we choose the two most commonly used forms of temperature variables, namely lagged hourly temperatures (T_{t-h}, $h = 1, 2, \ldots, 72$) and 24-hour (or daily) moving average temperatures, where the daily moving average temperature of the dth day can be written as:

$$\tilde{T}_{t,d} = \frac{1}{24} \sum_{h=24d-23}^{24d} T_{t-h}, \quad d = 1, 2, \ldots, 7. \quad (3)$$

Then, Tao's vanilla benchmark model can be extended to:

$$y_t = \beta_0 + \beta_1 \, Trend + \beta_2 M_t + \beta_3 W_t + \beta_4 H_t + \beta_5 W_t H_t + f(T_t) + \underbrace{\sum_d f(\tilde{T}_{(t,d)}) + \sum_h f(T_{t-h})}_{recency\ effect\ terms}. \quad (4)$$

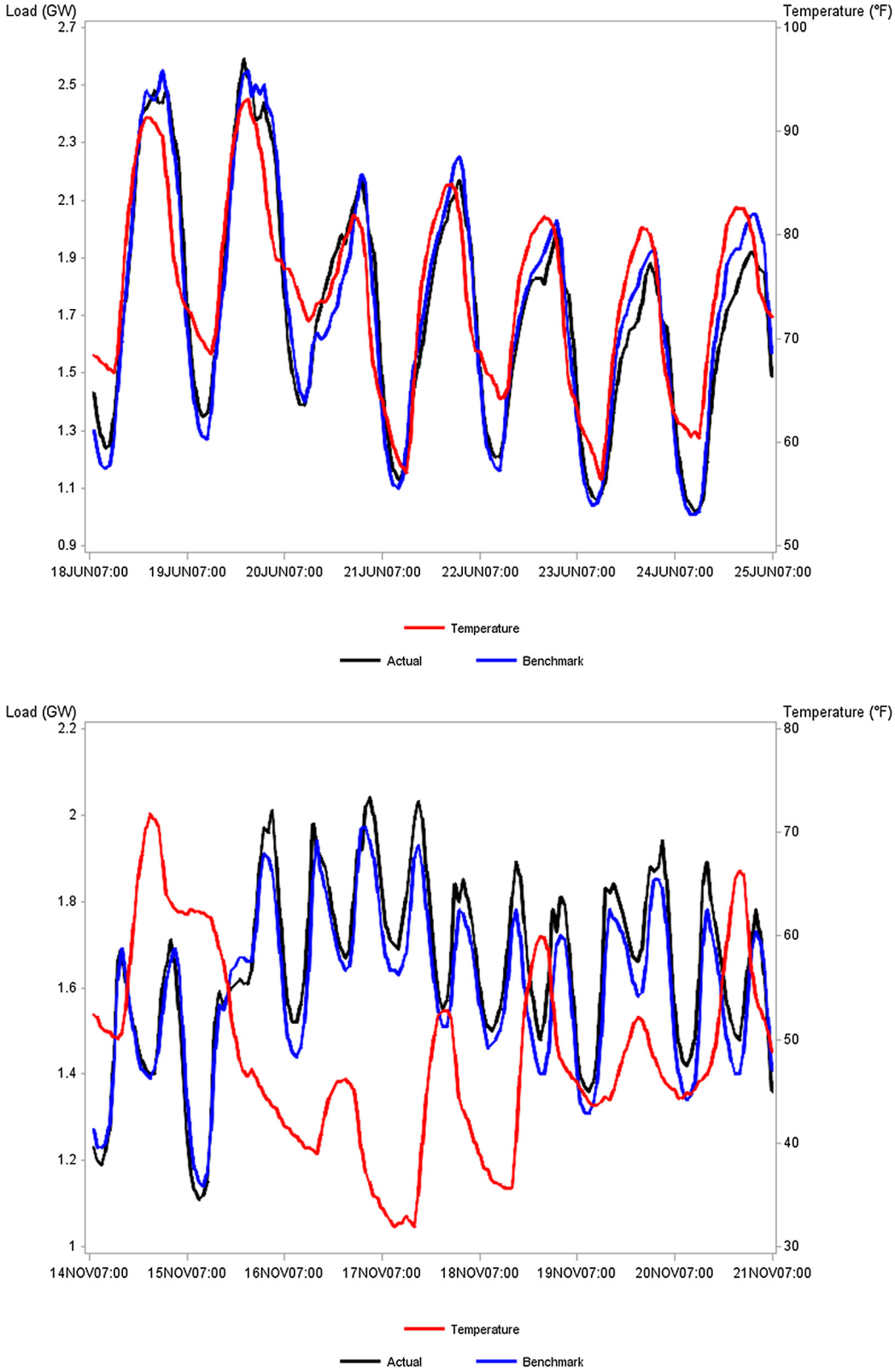

Fig. 2. Actual load, forecasted load and temperature plots for a summer week (upper) and a winter week (lower).

$h \setminus d$	0	1	2	3	4	5	6	7
0	4.89	4.10	4.09	4.16	4.23	4.13	4.12	4.25
1	4.55	3.92	3.91	3.97	4.05	3.95	3.96	4.08
2	4.34	3.81	3.80	3.87	3.94	3.85	3.85	3.97
3	4.20	3.76	3.74	3.81	3.88	3.79	3.79	3.91
4	4.09	3.71	3.70	3.76	3.84	3.75	3.74	3.86
5	4.00	3.68	3.67	3.73	3.81	3.72	3.71	3.83
6	3.93	3.65	3.64	3.71	3.78	3.70	3.69	3.81
7	3.86	3.63	3.62	3.69	3.76	3.68	3.67	3.80
8	3.81	3.60	3.60	3.67	3.75	3.67	3.66	3.78
9	3.77	3.59	3.58	3.65	3.73	3.66	3.65	3.78
10	3.74	3.58	3.57	3.64	3.72	3.65	3.64	3.77
11	3.73	3.57	3.55	3.63	3.71	3.64	3.63	3.76
12	3.71	3.56	**3.54**	3.62	3.69	3.63	3.62	3.75
13	3.69	3.56	3.54	3.62	3.68	3.62	3.62	3.74
14	3.67	3.57	3.55	3.63	3.69	3.63	3.63	3.76
15	3.66	3.58	3.57	3.64	3.70	3.64	3.64	3.77
16	3.67	3.60	3.58	3.66	3.71	3.66	3.66	3.79
17	3.67	3.62	3.61	3.68	3.73	3.68	3.68	3.81
18	3.67	3.64	3.63	3.71	3.75	3.70	3.70	3.83
19	3.68	3.67	3.65	3.73	3.77	3.72	3.71	3.86
20	3.68	3.69	3.68	3.75	3.79	3.74	3.73	3.88
21	3.69	3.71	3.70	3.77	3.80	3.76	3.76	3.90
22	3.70	3.73	3.72	3.78	3.81	3.78	3.77	3.92
23	3.72	3.74	3.73	3.78	3.82	3.79	3.79	3.94
24	3.73	3.76	3.75	3.79	3.83	3.80	3.80	3.95
25	3.74	3.76	3.77	3.80	3.84	3.81	3.81	3.96
26	3.75	3.78	3.79	3.81	3.85	3.83	3.82	3.98
27	3.76	3.80	3.81	3.83	3.86	3.84	3.84	4.00
28	3.78	3.82	3.83	3.85	3.88	3.86	3.86	4.03
29	3.79	3.84	3.86	3.88	3.90	3.88	3.89	4.06
30	3.81	3.86	3.89	3.90	3.93	3.91	3.91	4.09
31	3.83	3.87	3.90	3.92	3.94	3.93	3.93	4.12
32	3.85	3.89	3.93	3.94	3.97	3.96	3.96	4.14
33	3.87	3.91	3.95	3.97	3.99	3.99	3.99	4.18
34	3.90	3.94	3.98	3.99	4.02	4.02	4.02	4.21
35	3.93	3.97	4.00	4.01	4.03	4.04	4.04	4.24
36	3.94	3.98	4.02	4.03	4.05	4.06	4.06	4.27
37	3.96	3.99	4.05	4.05	4.08	4.08	4.09	4.29
38	3.97	4.00	4.07	4.08	4.11	4.12	4.13	4.33
39	3.99	4.02	4.10	4.10	4.13	4.15	4.16	4.36
40	4.01	4.03	4.12	4.12	4.16	4.17	4.18	4.39
41	4.04	4.05	4.14	4.14	4.18	4.19	4.20	4.42
42	4.07	4.07	4.16	4.15	4.20	4.22	4.23	4.45
43	4.09	4.09	4.17	4.17	4.23	4.24	4.26	4.48
44	4.11	4.11	4.18	4.18	4.25	4.26	4.28	4.51
45	4.13	4.12	4.19	4.20	4.26	4.27	4.30	4.53
46	4.14	4.13	4.21	4.21	4.28	4.29	4.33	4.55
47	4.15	4.14	4.22	4.22	4.30	4.31	4.35	4.58
48	4.16	4.16	4.24	4.24	4.33	4.33	4.38	4.60

Fig. 3. Heat map of MAPE values (in %) for recency effect modeling based on the validation data (the year 2006) for Z_{21}.

We model the recency effect using a trial-and-error method, varying the number of days from 0 to 7, and the number of lags from 0 to 72. In total, there are 584 (73×8) possible "average-lag" (or d–h) pairs.

In this paper, we again use the mean absolute percentage error (MAPE) of hourly loads as the error measure for evaluating load forecasts:

$$MAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{A_t - F_t}{A_t}\right|, \tag{5}$$

where A_t is the actual value and F_t is the forecasted value at time t.

Enumerating all 584 models based on different d–h pairs for Z_{21}, we can obtain the heat map corresponding to the MAPE values on the validation data (the year 2006), as shown in Fig. 3, which shows the first 48 lags to avoid a verbose presentation. The lowest MAPE is 3.54%, or, more precisely, 3.542%, which occurs in the model with two daily moving average temperatures and 12 lagged hourly temperatures.

Using this model ($d = 2$; $h = 4$) to perform one-year-ahead ex post forecasting on the test data (2007), with the previous two years (2005–2006) for parameter estimation, we get a MAPE value of 4.27%. Comparing this with the MAPE of 5.22% given by Hong et al. (2015) for Tao's vanilla benchmark model on the test data, the recency effect model reduces the MAPE value by 18%.

Table 1
MAPE values (in %) of the recency effect vs. benchmark models on 21 zones of the GEFCom2012 load forecasting data (one-year ahead ex post forecasting for 2007). Z_4 experienced a major outage, while Z_9 is an industrial customer.

	Zone	Recency	Benchmark
Aggregated zone	21	**4.27**	5.22
Regular zones	1	**5.80**	7.01
	2	**4.90**	5.62
	3	**4.90**	5.62
	5	10.71	**9.88**
	6	**4.90**	5.55
	7	**4.90**	5.62
	8	**6.66**	7.50
	10	**5.62**	6.70
	11	**6.53**	7.70
	12	**5.54**	6.78
	13	**6.81**	7.39
	14	**8.13**	9.38
	15	**6.63**	7.44
	16	**6.80**	8.12
	17	**4.32**	5.26
	18	**5.50**	6.72
	19	**6.68**	7.90
	20	**5.02**	5.74
	Average (regular zones)	**6.13**	7.00
Special zones	4	16.13	**16.08**
	9	140.9	**139.16**

Fig. 4 shows a comparison of the forecasts from the benchmark and recency effect models for the weeks shown in Fig. 2. We observe that (1) the recency effect model captures peaks and troughs much more closely than the benchmark model; and (2) the leading/lagging issue of the benchmark model has been resolved by including the recency effect. Overall, the model with the recency effect captures more salient features than Tao's vanilla benchmark model.

3.2. In a geographic hierarchy

The geographic hierarchy in hierarchical load forecasting refers to the households, small areas, sub-regions, and regions, rolling up to the top aggregated level. The GEFCom2012 dataset itself has 20 zones at the bottom level, with Z_{21} as the sum of those 20 zones. In this section, we apply the recency effect modeling to all 21 load zones in the GEFCom2012 dataset.

The GEFCom2012 data does not include information as to locations. To obtain a rough estimate of the locations, we first take the annual average temperature of each virtual weather station, then sort the average temperatures in ascending order, to represent zones from north to south. Fig. 5 shows the d–h pairs for the 19 zones (18 regular zones plus Z_{21}) sorted by the average annual temperature. Overall, the recency effect appears to differ across most of the 19 zones, due to the fact that different zones will have different compositions of residential, commercial and industrial customers, which may combine to have different responses to weather patterns.

Table 1 lists the MAPE values for one-year-ahead ex post forecasts of the year 2007, using the previous two years (2005–2006) for parameter estimation. As was discussed by Hong et al. (2015), Z_4 experienced a major outage, and Z_9 is an industrial customer. Therefore, these two special zones are listed separately from the others. At the aggregated level, modeling the recency effect helps to reduce the MAPE value of the benchmark model by 18%. At the bottom level, the forecasting error can be improved for 17 of the 18 regular zones when modeling the recency effect in a customized fashion. On average, the MAPE values are 12% lower for the recency effect models than for the benchmark model. The improvement in forecast accuracy through the customization of each low-level zone is also consistent with the conclusion of Hong et al. (2015) that different zones may need different sets of weather stations to enhance the forecast accuracy.

3.3. By hour of a day

The hourly electricity demand series naturally has multiple seasonal patterns, such as hours of a day, days of a week, months of a year, and so forth. These calendar variables automatically form the temporal hierarchy. In the load forecasting literature, some researchers have forecasted the load based on the methodology of using all of the historical data to derive one interaction regression model (Hong, 2010; Hong & Wang, 2014; Hong, Wilson, & Xie, 2014b). Many other researchers have sliced the data into 24 pieces, so that the parameter estimation of the load forecasting model is conducted individually for each of the 24 groups (Ben Taieb & Hyndman, 2014; Black & Henson, 2014; Fan & Hyndman, 2012; Ramanathan, Engle, Granger, Vahid-Araghi, & Brace, 1997). There has not been any formal comparison of the two. In addition, there has not been any formal study exploring the possibility of modeling each hour with a different variable combination, though (Ramanathan et al., 1997) mentioned briefly that different variable combinations were used to model different hours. In this section, we test four combinations of cross validation and model settings, as shown in Table 2.

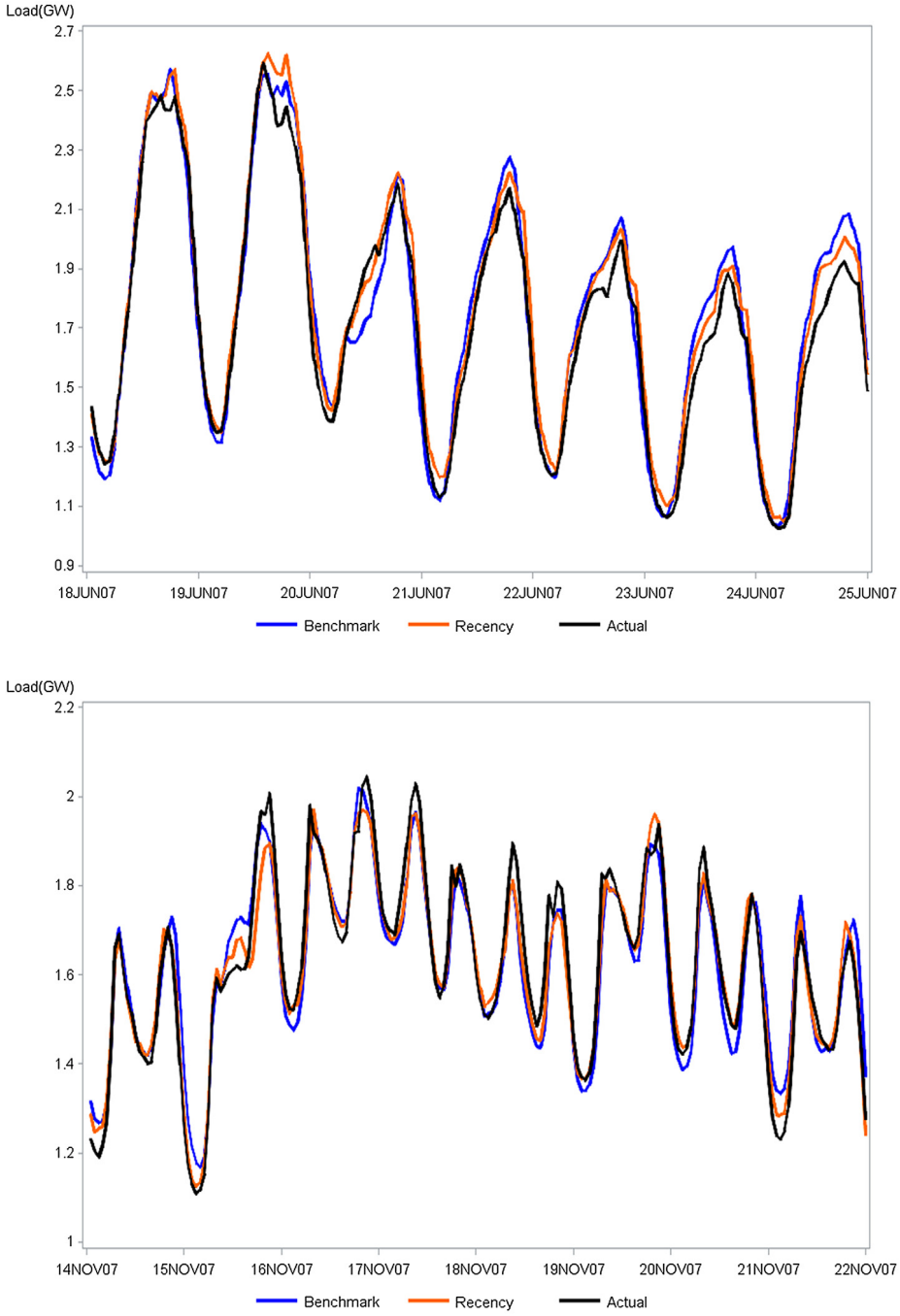

Fig. 4. Actual load, forecasted load from the benchmark model, and forecasted load from the recency effect model for a summer week (upper) and a winter week (lower).

A. This is our base case, and is conducted in Section 3, where we identify the d–h pair through one training dataset and one validation dataset. This implementation results in the same variables (a.k.a. one d–h pair) and parameters being used for all 24 hours.
B. We first slice the data into 24 pieces, by hour of the day. Each piece includes a training set and a validation set. We estimate a model for each hour of the day using the training set for the corresponding hour, then calculate the MAPE on the validation set. The d–h pair that results in the lowest average MAPE on the 24 validation sets is selected as the pair for the 24 models. This implementation also results in the same variables being used for all 24 hours, but the parameters are estimated separately for each hour of a day. In total, we have 24 sets of parameters, one for each hour.
C. We slice the validation data into 24 pieces by hour of the day, then estimate a model using the full (i.e., all hours)

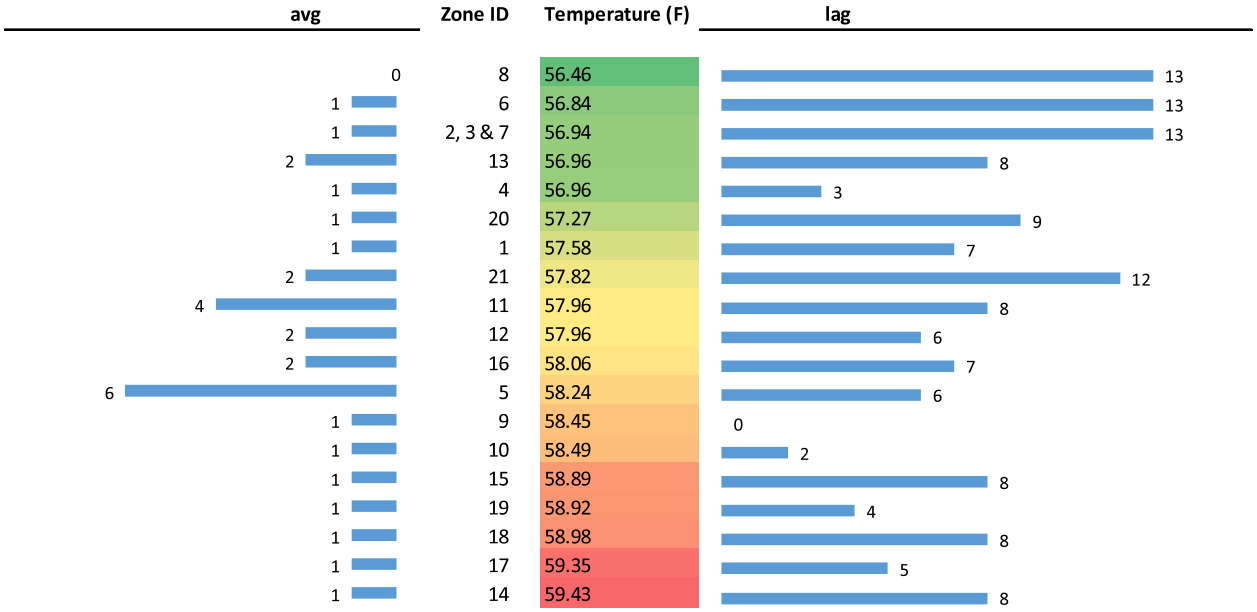

Fig. 5. Recency effects (in *d*–*h* pairs) for the regular zones, sorted by annual average temperature.

Table 2
Different implementations of recency effect modeling.

	Training	Validation	*d*–*h* pair
A	1	1	1
B	24	24	1
C	1	24	24
D	24	24	24

training set. The *d*–*h* pair that results in the lowest MAPE on each validation set is selected as the pair for the corresponding hour of the day. In total, we select 24 *d*–*h* pairs. In other words, this implementation results in 24 sets of variables and parameters, one for each hour of a day.

D. The data processing is similar to that of Option B. We first slice the data into 24 pieces by hour of the day. Each piece includes a training set and a validation set. However, instead of selecting a common *d*–*h* pair for all 24 models, we designate one *d*–*h* pair for each model. In total, we select 24 *d*–*h* pairs. This implementation results in 24 sets of variables and parameters, one for each hour of a day.

Based on the above four implementation options, we can develop four models, of which the parameters (values of *d* and *h*) are shown in the radar charts in Fig. 6. Since options A and B involve using one *d*–*h* pair across all 24 hours of a day, the radar charts show their territories as circles. When removing this constraint of having the same *d*–*h* pair for all hours, different hours of a day may present different recency effects. This observation is similar to that made in Table 1, namely that different zones may present different recency effects, which is again due to the fact that the relationship between electricity consumption and the temperature varies through the day. Nevertheless, we can observe similar or identical *d*–*h* pairs in some adjacent hours. Fig. 6 also shows that the territories of options B and D are significantly less than those of options A and C. In other words, options B and D include fewer variables than options A and C. This is because options B and D estimate models using only 1/24 of the observations used by options A and C. The number of observations in the training data limits the number of variables that a model can hold.

We then apply these four models to one-year-ahead ex post forecasting of the loads for 2007. The error statistics from these four options are listed in Table 3. On average, option A leads to the lowest MAPE values. In other words, in this case study, identifying recency effect modeling using one training dataset and one validation dataset outperforms the three alternative implementations.

Note that the common practice in the 1990s and 2000s has been to use one set of variables for all hours of a day, which makes computation easy. Ramanathan et al. (1997) provided one of the first papers to report customized models for different hours of a day. In this paper, we advance the state-of-the-art further by taking advantage of modern computing power to remove the limit on the number of variables being included in a regression model, leading to a superior accuracy.

Furthermore, this part of the case study also provides empirical evidence to show that slicing the data into 24 pieces in order to develop one model for each hour, as many papers in the load forecasting literature have done, is not necessarily better than the other three alternatives listed in Table 2. On the other hand, although option A has the lowest MAPE value in this case study, it may or may not be the best one in other case studies with different datasets. When slicing the data into multiple pieces or modeling each hour individually, we are reducing the noise for each group, but also reducing the signal that could have been taken advantage of from the other groups. In other words, we have to find the balance between signal and noise, which is unknown and must be found by trial-and-error. This answer can vary on different data sets.

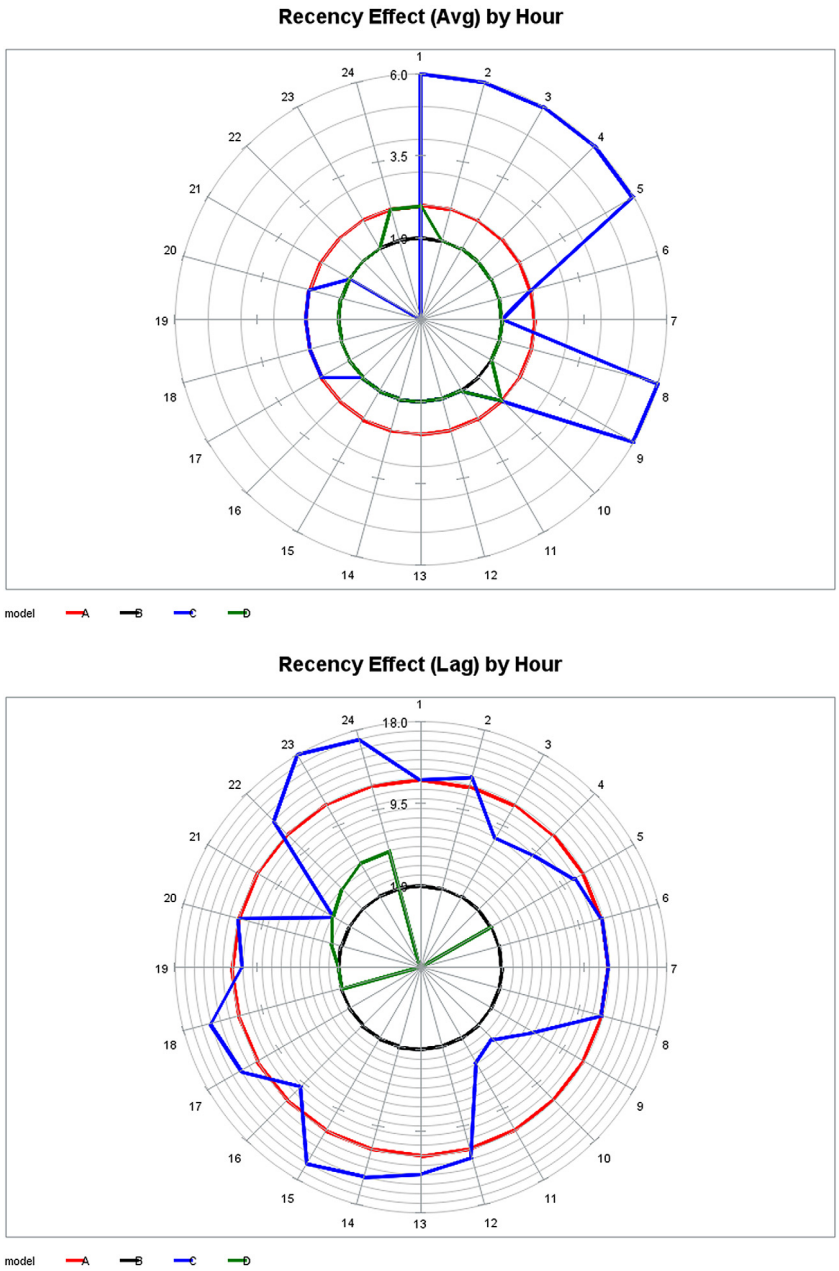

Fig. 6. The average-lag parameters for the four options A–D.

4. Discussions

4.1. More comparisons

So far in this paper, all of the recency effect models have been compared with Tao's vanilla benchmark model using the weather stations selected by Hong et al. (2015). The forecast horizon has been set to one year. In this section, we will expand the comparative study to other models in the setting of 24-hour-ahead ex post forecasting for the year 2007. The forecasts of hours 1–24 are issued at midnight each day on a rolling basis, using a 730-day moving window for parameter estimation.

Here, we introduce the four extra models for further comparisons:

(1) An advanced model (Adv_1) modified from the recency effect models proposed by Hong (2010), where we let $d = 1$ and $h = 3$ for all hours. This is to see whether stretching the numbers of average and lagged temperatures can help to improve the accuracy relative to models developed with computational constrains.

(2) An advanced model (Adv_2) simplified from the models in Ben Taieb and Hyndman (2014). Here, we plug the temperature variables identified by Ben Taieb and Hyndman (2014) into Eq. (2), with each hour

Table 3
MAPE values (in %) of recency effect models by hour of a day (one-year-ahead ex post forecasting for 2007).

Hour	A	B	C	D	Benchmark
1	**4.37**	4.50	4.77	4.68	5.82
2	4.56	4.59	4.97	**4.32**	5.84
3	4.59	4.28	5.24	**4.27**	5.82
4	4.69	4.47	5.22	**4.23**	5.71
5	4.61	4.41	5.07	**4.41**	5.75
6	4.66	**4.50**	4.66	4.52	5.70
7	4.86	4.93	**4.78**	4.81	5.56
8	4.28	4.21	4.70	**4.19**	4.85
9	3.71	**3.64**	4.00	3.70	4.32
10	**3.39**	3.61	3.46	3.75	4.16
11	3.53	3.98	**3.48**	3.81	4.25
12	**3.75**	4.18	3.79	4.07	4.51
13	**4.00**	4.45	4.02	4.32	4.52
14	**4.20**	4.54	4.22	4.57	4.68
15	**4.50**	4.62	4.54	4.80	4.82
16	**4.61**	4.73	4.67	4.82	5.02
17	4.65	4.70	**4.64**	4.81	5.12
18	4.70	4.72	4.68	**4.72**	5.39
19	4.33	4.27	4.31	**4.27**	5.52
20	**4.22**	4.34	4.22	4.23	5.75
21	4.14	4.19	**4.07**	4.20	5.36
22	3.90	4.27	**3.81**	4.02	5.56
23	4.00	4.48	**3.82**	4.46	5.53
24	4.30	4.49	**4.06**	4.95	5.73
Average	**4.27**	4.38	4.38	4.37	5.22

modeled as a separate group. Note that this is not a direct reproduction of the models of Ben Taieb and Hyndman (2014), because this paper does not include lagged loads, as was mentioned in Section 2.2. Moreover, the model structure we are using in this paper differs from those of Ben Taieb and Hyndman (2014). The analogy of integration discussed by Hong and Fan (2016) suggests that a subset of the variables selected by Ben Taieb and Hyndman (2014) for each hour may not be the optimal selection for the model structure here. The purpose of this comparison is to have a sanity check as to whether the recency effect models proposed in this paper perform worse than a simplified version of the models from a top-five team from GEFCom2012.

(3) A daily seasonal naïve benchmark (SNM-D). It uses the load of hour $t-24$ as the forecast for the load of hour t.
(4) A weekly seasonal naïve benchmark (SNM-W). It uses the load of hour $t-168$ as the forecast for the load of hour t.

Table 4 lists the 24-hour-ahead ex post forecasting MAPE values for all 21 zones from the recency effect model (option A), two advanced models (Adv_1 and Adv_2), the benchmark model, and two seasonal naïve models. At the aggregated level (Z_{21}), modeling the recency effect reduces the MAPE value of the benchmark model by 21% for 24-hour-ahead rolling forecasting. At the bottom level, the recency effect models (option A) win for 12 of the 18 regular zones, reducing the MAPE values of the benchmark model by 15% on average. The Adv_1 model wins the other 6 regular zones, with the average MAPE value slightly higher than the recencey effect models of option A.

Both seasonal naïve models perform very poorly compared with the other four models. Seasonal naïve models are used commonly for benchmarking purposes in other industries, such as the retail and manufacturing industries. In load forecasting, the two applications in which seasonal naïve models are most useful are: (1) benchmarking the forecast accuracy for very unpredictable loads, such as household level loads; and (2) comparisons with univariate models. In most other applications, however, the seasonal naïve models and other similar naïve models are not very meaningful, due to the lack of accuracy.

Table 5 lists the MAPE values of the recency effect model (option A), two advanced models mentioned above, and the benchmark model. This time, we zoom into the error statistics by hour. To avoid verbose presentation, we only

Table 4
MAPE values (in %) of the recency effect vs. benchmark models on all 21 zones of the GEFCom2012 load forecasting data (24-hour-ahead ex post forecasting for 2007). Z_4 experienced a major outage, while Z_9 is an industrial customer.

	Zone	Recency (A)	Adv_1	Adv_2	Benchmark	SNM-D	SNM-W
Aggregated zone	21	**3.86**	3.97	4.45	4.88	8.40	14.30
Regular zones	1	**5.62**	5.82	6.83	7.00	11.28	19.01
	2	**4.59**	4.65	5.36	5.25	8.55	12.57
	3	**4.59**	4.65	5.36	5.25	8.55	12.57
	5	7.69	**7.28**	8.62	8.32	12.59	20.47
	6	**4.57**	4.65	5.29	5.27	8.55	12.79
	7	**4.59**	4.65	5.36	5.25	8.55	12.57
	8	**6.02**	6.26	7.24	6.85	9.40	15.96
	10	**4.97**	**4.97**	6.17	5.85	12.23	15.80
	11	**5.71**	5.81	6.38	6.88	10.77	18.13
	12	5.11	**5.09**	5.83	6.41	11.94	20.31
	13	**6.31**	6.36	7.60	7.13	9.01	15.23
	14	7.60	**7.56**	9.12	9.10	15.66	25.14
	15	6.29	**6.26**	7.34	7.19	10.56	17.69
	16	6.37	**6.31**	7.83	7.69	14.12	23.75
	17	**4.13**	4.14	4.94	5.09	10.58	15.65
	18	**5.25**	5.29	6.26	6.50	11.58	19.51
	19	**6.14**	6.16	7.50	7.60	13.12	21.77
	20	**4.54**	4.62	5.38	5.20	8.15	12.72
	Average	**5.56**	5.59	6.58	6.55	10.84	17.31
Special zones	4	**14.51**	**14.51**	15.97	14.74	15.52	20.73
	9	109.84	107.21	115.18	113.79	47.53	120.06

Table 5
MAPE value (in %) comparison of the recency model (option A), two advanced models and the benchmark by hour of a day (24-hour-ahead ex post forecasting for 2007).

Hour	Recency (A)	Adv-1	Adv-2	Benchmark
1	**3.73**	4.26	4.57	5.41
2	**3.83**	4.22	4.67	5.33
3	**3.88**	4.24	4.42	5.33
4	**3.98**	4.18	4.81	5.28
5	**3.95**	4.11	4.24	5.33
6	**4.13**	4.14	4.25	5.24
7	4.53	**4.33**	5.04	5.18
8	4.08	**3.86**	4.51	4.58
9	3.51	**3.36**	3.61	3.97
10	3.16	**3.11**	3.20	3.74
11	**3.30**	3.33	3.55	3.87
12	**3.51**	3.56	4.07	4.20
13	**3.59**	3.69	4.49	4.33
14	**3.76**	4.00	4.68	4.52
15	**4.00**	4.22	4.99	4.62
16	**4.12**	4.30	5.63	4.73
17	**4.23**	4.26	5.09	4.72
18	**4.39**	4.51	4.95	5.05
19	**4.06**	4.10	4.31	5.37
20	3.93	**3.87**	4.50	5.37
21	3.66	**3.63**	4.22	4.91
22	**3.57**	3.68	3.86	5.15
23	**3.73**	4.11	4.42	5.32
24	**3.97**	4.28	4.64	5.47
Average	**3.86**	3.97	4.45	4.88

show the results from Z_{21}, because the results from the other zones do not provide additional information. Overall, the recency effect model wins for 18 of the 24 hours in a day, while the Adv_1 model wins the other six. On average, the recency effect model is the most accurate of the four, followed by the Adv_1 model, and then the Adv_2 model.

4.2. Big data in electric load forecasting

The utilities have been working with big data ever since it was called just "data". This smart grid era has witnessed the growth of data into big data. Nowadays, the term "big data" is no longer new to the utility industry. In fact, it is gradually becoming a buzzword in both the industry and academia. With the term "big data" in the title of this paper, we would like to expand the discussion a bit to cover the concept of big data in load forecasting, or energy data analysis in general.

Fundamentally, the "big" in big data should be relative to what has happened in the past. Therefore, we define the criteria for big data in relative terms:

(1) The data size is larger than typical data analysis tools (including the IT infrastructure around the tools) can handle. For instance, if MS Excel 2003 is the tool used for data analysis, then a data file with 5 million rows is big. If the data should come in at 1 GB/s, but the internet downloading speed is 10 MB/s, then the data stream is big.
(2) The computing time is longer than the analysis time or lead time for the relevant business applications. If it takes a few hours to think of the design of an algorithm, but a few weeks to test the algorithm, then it is big data problem. If it takes a few hours to calibrate an hour-ahead forecasting model, which should be re-calibrated every hour, then it is a big data problem.
(3) The problem requires analysis at a higher level of granularity than usual. If the typical load forecasting process relies on monthly data, moving to daily or hourly data may introduce the big data challenge.

Although these three criteria do not all have to be met at the same time in order for a problem to qualify as a big data problem, they are indeed connected with each other. The analysis of high resolution data with the customized models pushed down to low levels often requires the use of advanced data analysis tools, and takes significant computing time.

The title of this paper includes big data because it covers the last two criteria. The regression models that we developed in this paper contain up to thousands of variables. This requires a significant amount of time for parameter estimation, much longer than for our thought process. Moreover, we have customized the models based on the individual zones of a geographic hierarchy and the individual nodes (hours) of the temporal hierarchy. The forecasting error reduction achieved through our proposed approach also suggests that the use of high performance computing facilities for load forecasting may make sense in some circumstances.

The fact that the recency effect model beats the Adv_1 model confirms the effectiveness of taking a big data approach to improving forecast accuracy. Nevertheless, we can also observe the diminishing improvement as the customized modeling effort is extended to the low level of the geographic hierarchy, which indicates that such an approach may have limitations. In other words, we may not want to apply these large models down to the very low level of power systems. This also makes sense, because the load is much more volatile and unpredictable at very low levels, such as the household level, than at aggregated levels. Additional research is needed to allow the appropriate level to be identified, in order to conduct customized modeling efforts.

In terms of forecasting techniques, this paper has applied multiple linear regression with an ordinary least squares estimator, due mainly to its simplicity and practicality. Nevertheless, other techniques, such as ridge regression, Bayesian regression or dynamic factor models, could also be plugged into our solution framework. The key idea of this paper relates to the customized modeling of the recency effect. Regression analysis is just one of many techniques that could be used to convey the same message.

4.3. Future research directions

This paper not only answers the fundamental question of how many average and lagged temperatures are needed in the model, but also opens a wide variety of research topics for future work:

(1) The use of cross validation methods to achieve an optimal recency effect. We show the 584 d–h pairs from Fig. 2 again, and draw another heat map for the test data (the year 2007) for Z_{21}, using data from 2005 and 2006 to estimate the parameters. As Fig. 7

Lag/d	0	1	2	3	4	5	6	7
0	5.22	4.51	4.57	4.60	4.64	4.80	4.88	4.98
1	4.94	4.40	4.48	4.52	4.57	4.75	4.86	4.93
2	4.75	4.33	4.41	4.45	4.50	4.68	4.79	4.86
3	4.60	4.29	4.37	4.40	4.46	4.64	4.75	4.82
4	4.49	4.25	4.34	4.36	4.43	4.60	4.71	4.78
5	4.40	4.24	4.32	4.34	4.40	4.57	4.68	4.74
6	4.34	4.23	4.31	4.32	4.39	4.55	4.66	4.72
7	4.28	4.24	4.30	4.32	4.39	4.54	4.65	4.71
8	4.25	4.25	4.30	4.32	4.39	4.54	4.64	4.71
9	4.23	4.25	4.30	4.31	4.38	4.53	4.63	4.69
10	4.21	4.25	4.29	4.31	4.38	4.51	4.61	4.68
11	4.20	4.26	4.28	4.31	4.38	4.51	4.59	4.67
12	4.20	4.26	4.27	4.31	4.38	4.50	4.58	4.66
13	4.19	4.27	4.27	4.31	4.38	4.50	4.57	4.66
14	4.18	4.26	4.27	4.31	4.38	4.49	4.57	4.66
15	4.18	4.26	4.26	4.31	4.38	4.50	4.57	4.66
16	**4.17**	4.25	4.26	4.31	4.39	4.50	4.58	4.67
17	4.18	4.26	4.26	4.32	4.40	4.51	4.59	4.67
18	4.18	4.26	4.27	4.33	4.40	4.51	4.60	4.68
19	4.19	4.28	4.29	4.34	4.41	4.53	4.61	4.69
20	4.20	4.28	4.30	4.35	4.42	4.54	4.63	4.70
21	4.21	4.29	4.31	4.36	4.42	4.55	4.64	4.71
22	4.22	4.29	4.32	4.37	4.43	4.56	4.65	4.72
23	4.23	4.30	4.34	4.38	4.43	4.57	4.66	4.73
24	4.24	4.31	4.35	4.39	4.44	4.57	4.67	4.73
25	4.25	4.31	4.36	4.41	4.46	4.59	4.68	4.74
26	4.26	4.32	4.38	4.43	4.48	4.60	4.69	4.75
27	4.28	4.34	4.41	4.45	4.50	4.62	4.71	4.78
28	4.29	4.36	4.43	4.47	4.52	4.65	4.74	4.81
29	4.30	4.37	4.45	4.48	4.54	4.67	4.75	4.83
30	4.31	4.39	4.46	4.50	4.57	4.69	4.77	4.86
31	4.33	4.41	4.48	4.52	4.59	4.71	4.80	4.89
32	4.35	4.43	4.50	4.53	4.60	4.73	4.82	4.92
33	4.36	4.44	4.50	4.54	4.60	4.73	4.82	4.93
34	4.37	4.46	4.50	4.54	4.61	4.74	4.84	4.96
35	4.39	4.47	4.51	4.56	4.62	4.77	4.86	4.99
36	4.40	4.48	4.52	4.57	4.63	4.78	4.88	5.02
37	4.41	4.49	4.55	4.59	4.66	4.81	4.92	5.06
38	4.43	4.49	4.56	4.61	4.68	4.83	4.94	5.09
39	4.44	4.50	4.56	4.62	4.69	4.85	4.96	5.11
40	4.45	4.51	4.57	4.64	4.70	4.86	4.97	5.13
41	4.47	4.51	4.57	4.65	4.71	4.87	4.99	5.15
42	4.49	4.53	4.58	4.67	4.73	4.88	5.00	5.18
43	4.51	4.54	4.59	4.68	4.74	4.89	5.02	5.20
44	4.53	4.55	4.61	4.69	4.76	4.91	5.03	5.21
45	4.54	4.56	4.62	4.71	4.78	4.93	5.06	5.24
46	4.55	4.58	4.63	4.72	4.79	4.95	5.07	5.26
47	4.57	4.59	4.65	4.74	4.82	4.97	5.09	5.28
48	4.58	4.60	4.66	4.74	4.82	4.98	5.10	5.28

Fig. 7. Heat map of MAPE values (in %) for recency effect modeling based on test data (from 2007) for Z_{21}.

shows, the lowest MAPE value in hindsight is 4.17%, which occurs for $d = 0$ and $h = 16$. Since the future is stochastic by nature, the d–h pair previously selected from the validation data (the year of 2006), in this case $d = 2$ and $h = 12$, may not be the optimal d–h pair in hindsight. A similar phenomenon was observed and discussed by Hong et al. (2015) in regard to optimal weather station selection. In this paper, we selected models using a simple out-of-sample test. One possible future research direction could be the examination of different cross validation methods, such as those discussed by Arlot and Celisse (2010), for model selection.

(2) An enhancement of the computational efficiency. In this paper, we did not use any advanced search algorithms to identify the *d-lag* pair that resulted in the lowest validation error. Based on Fig. 3, it is clear that searching the whole space, i.e., 8×73,

may not be necessary. We could try to design some heuristic search algorithms for reducing the search space without compromising the forecasting accuracy.

(3) Dimension reduction. A model requiring the estimation of thousands of parameters may be larger than is necessary. We can investigate whether dimension reduction techniques can help to make less-verbose models, while reducing the forecast error. For instance, we could try combining some hours of a day, in order to reduce the number of parameters requiring estimation. A similar analogy was used by Hong (2010) for combining some weekdays. Similar ideas have also been applied to electricity price forecasting, as was discussed by Weron (2014).

(4) Forecast combination. This paper has generated many models with similar structures and levels of accuracy. Rather than selecting one model via cross validation, we could also combine the various forecasts. Liu, Nowotarski, Hong, and Weron (in press) called these similar models "sister models", with their forecasts being referred to as "sister forecasts". The forecast combination output can be either a probabilistic forecast, as was done by Liu et al. (in press), or another point forecast.

5. Conclusion

In conclusion, this paper investigates the recency effect for electric load forecasting, the fundamental relationship between the load and the temperatures of preceding hours. The idea is to determine the numbers of lagged hourly temperatures and daily moving average temperatures needed to enhance the load forecasting accuracy without considering the computing power constraints. The case study is based on the data published in GEFCom2012. We demonstrate the recency effect from three aspects: (1) the aggregated level (Z_{21}) of the geographic hierarchy; (2) the 20 zones at the bottom level of the geographic hierarchy; and (3) the 24 hours of a day. Overall, the recency effect model significantly outperforms the benchmark model based on the MAPE of the holdout sample (test data, the year 2007 in this paper). The recency effect appears to be different for each zone of the geographic hierarchy and each hour of the day. The case study has shown that using the proposed approach to customize the model for each zone or hour of the day can help to enhance the accuracy of the hierarchical load forecasts.

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